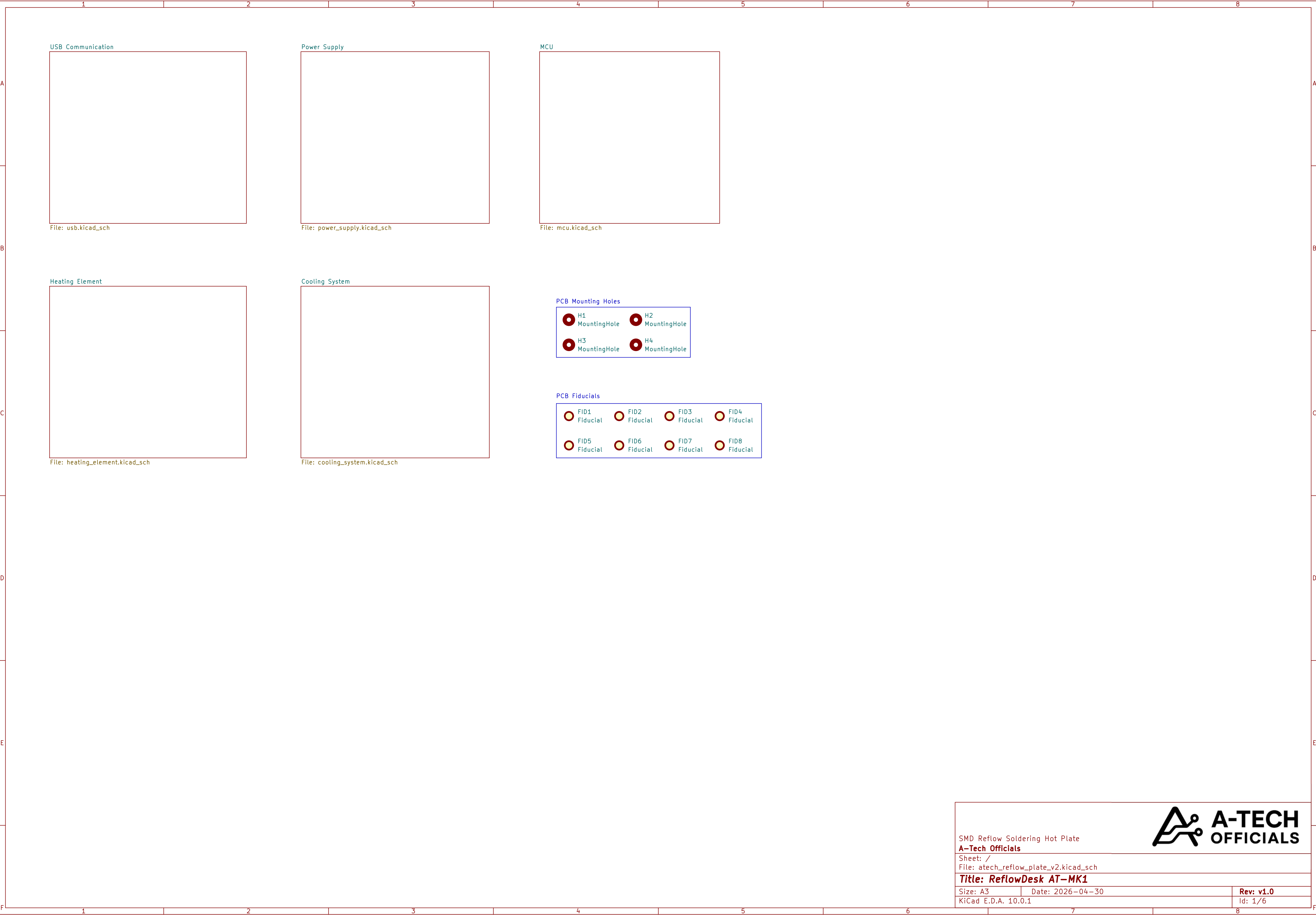
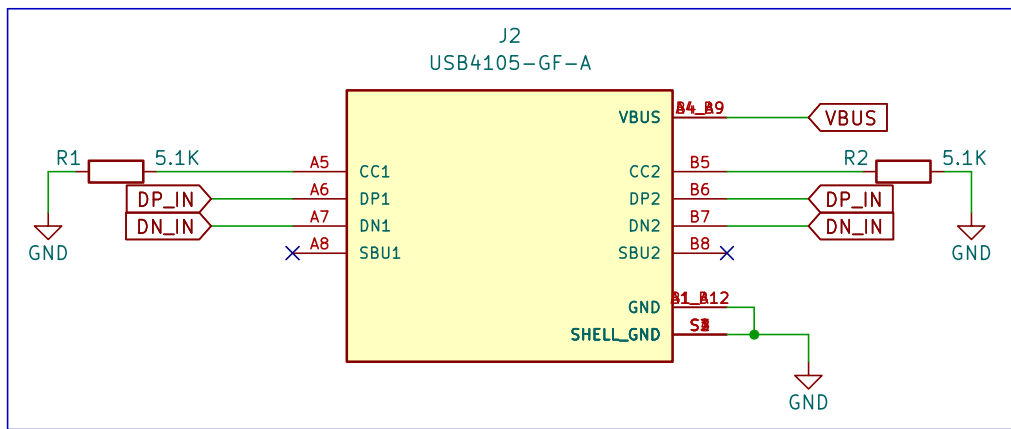
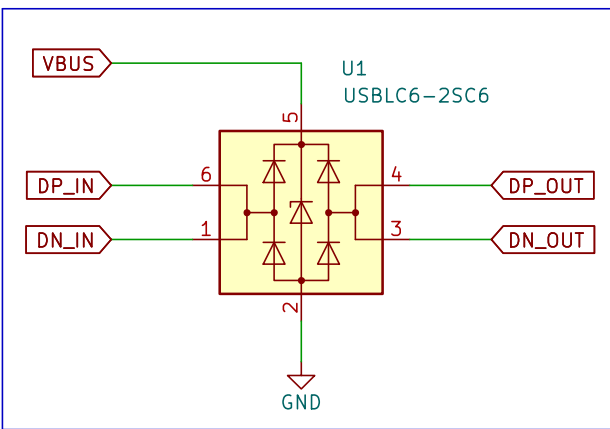




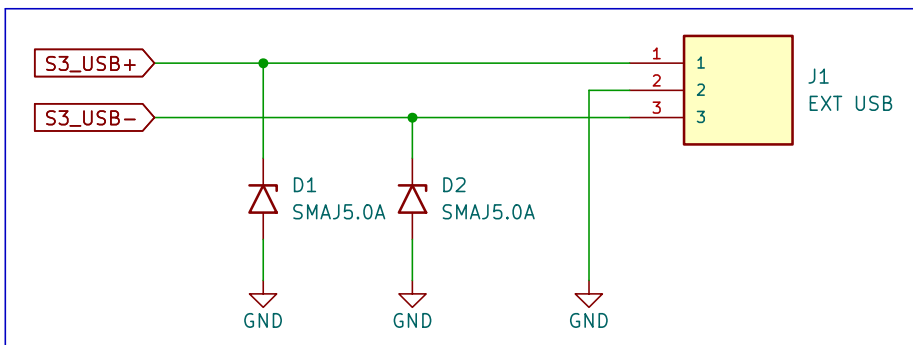
USB 3.1 Type-C Connector (Receptacle)



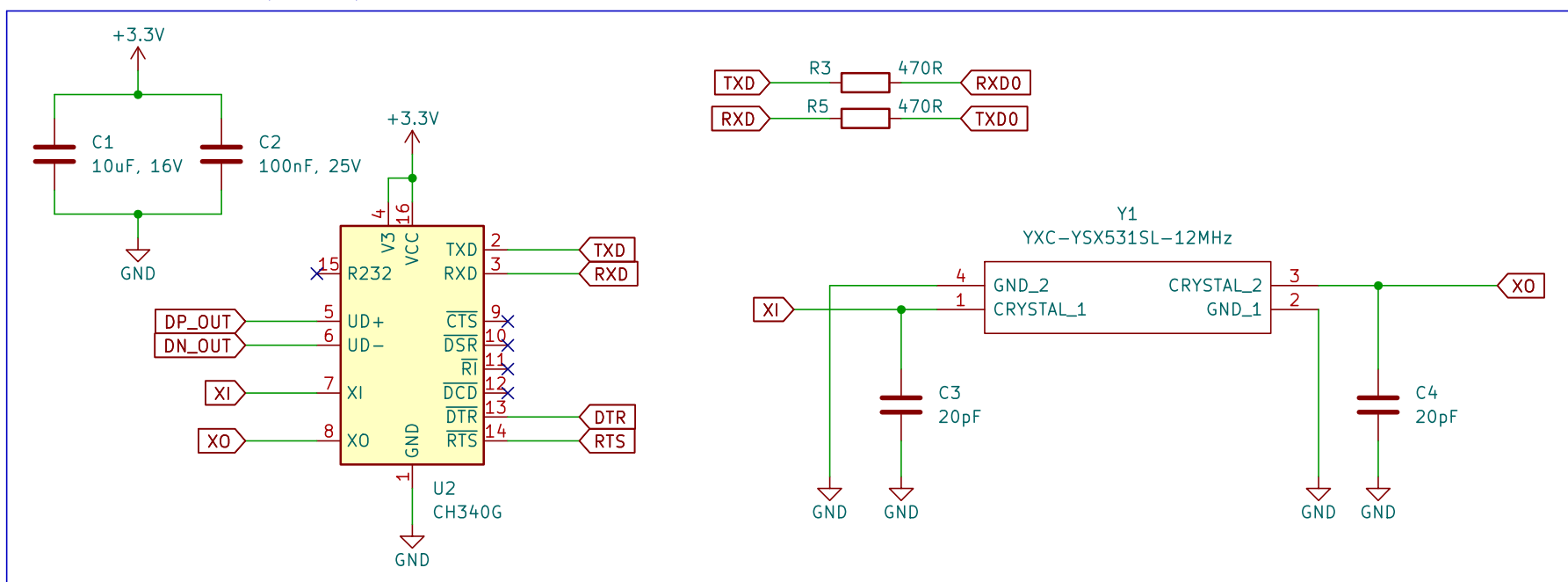
ESD Protection Circuit



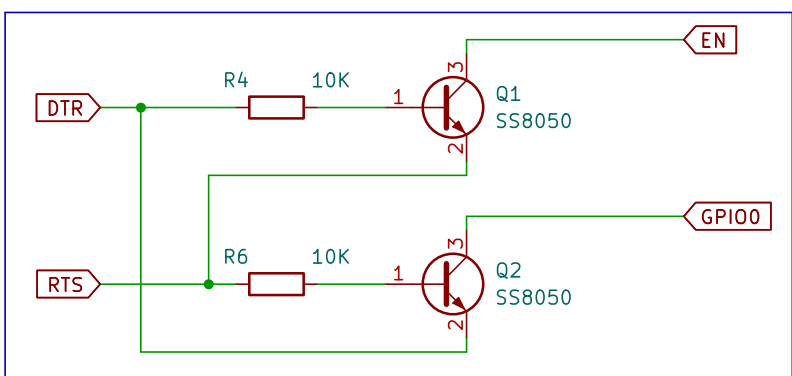
External USB Module Connector



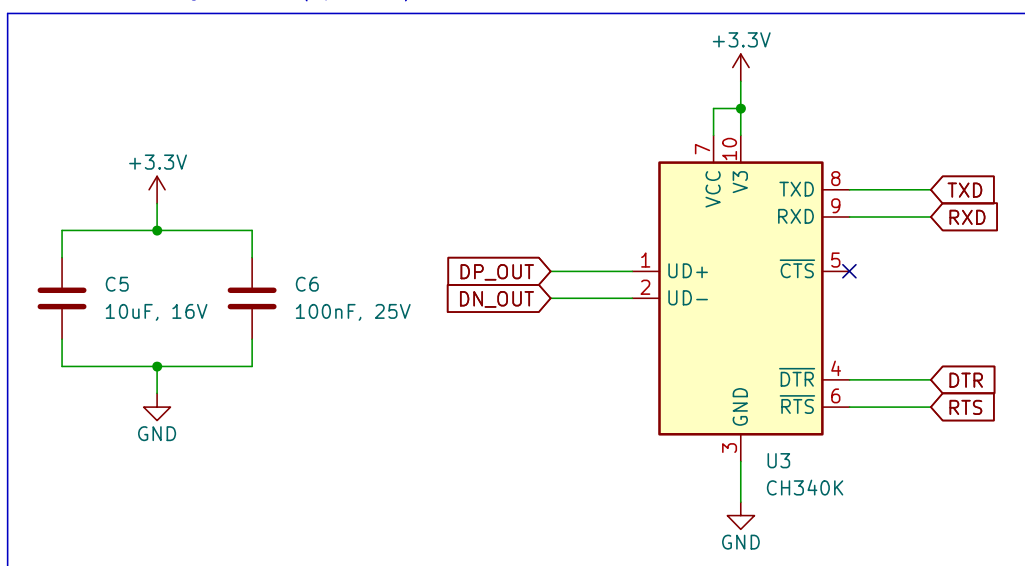
USB to UART Bridge Circuit (Option-2)



ESP32 MCU Auto Programming Circuit



USB to UART Bridge Circuit (Option-1)



SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /USB Communication/

File: usb.kicad_sch

Title: ReflowDesk AT-MK1 (USB Communication)

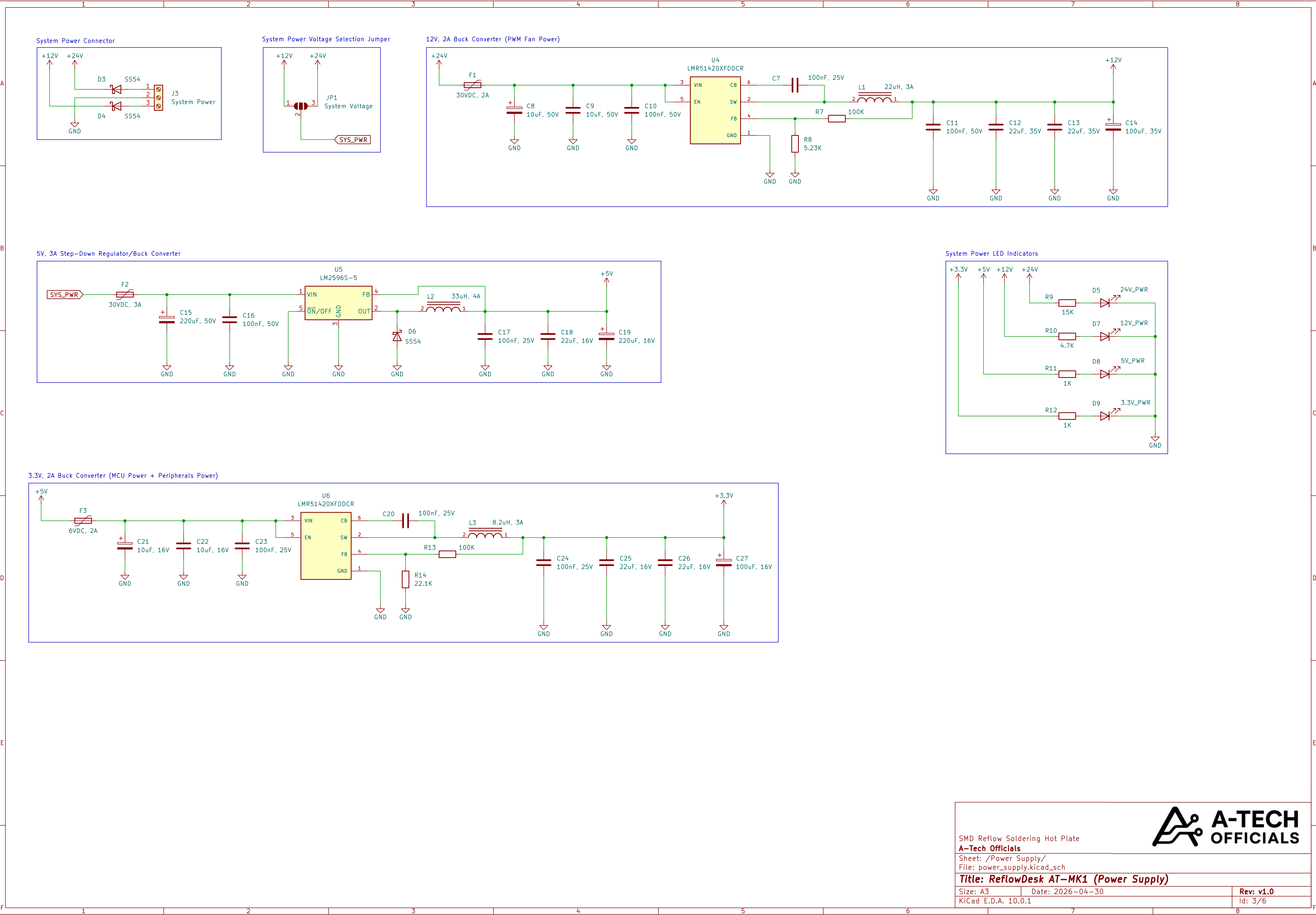
Size: A3

Date: 2026-04-30

Rev: v1.0

KiCad E.D.A. 10.0.1

Id: 2/6

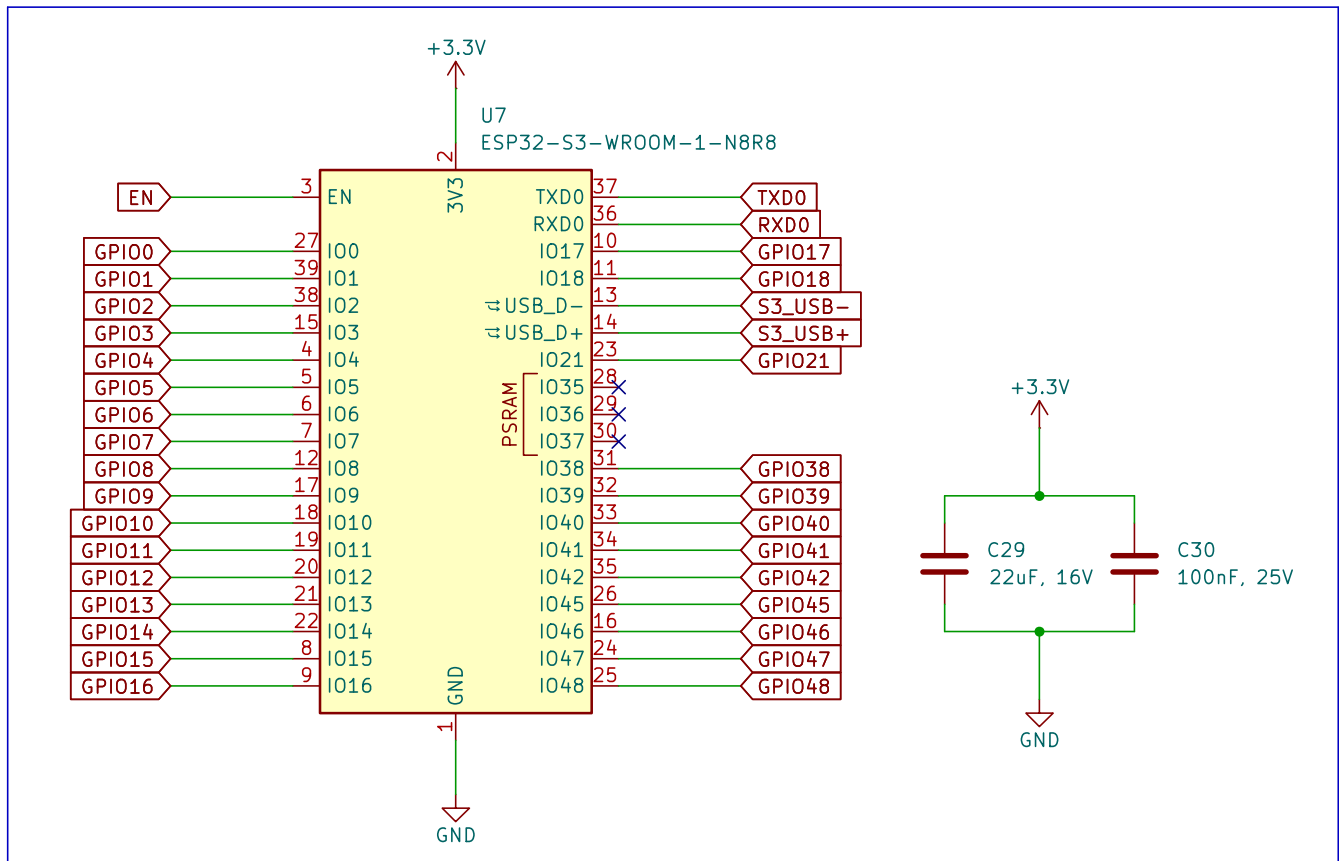


SMD Reflow Soldering Hot Plate
A-Tech Officials
Sheet: /Power Supply/
File: power_supply.kicad_sch

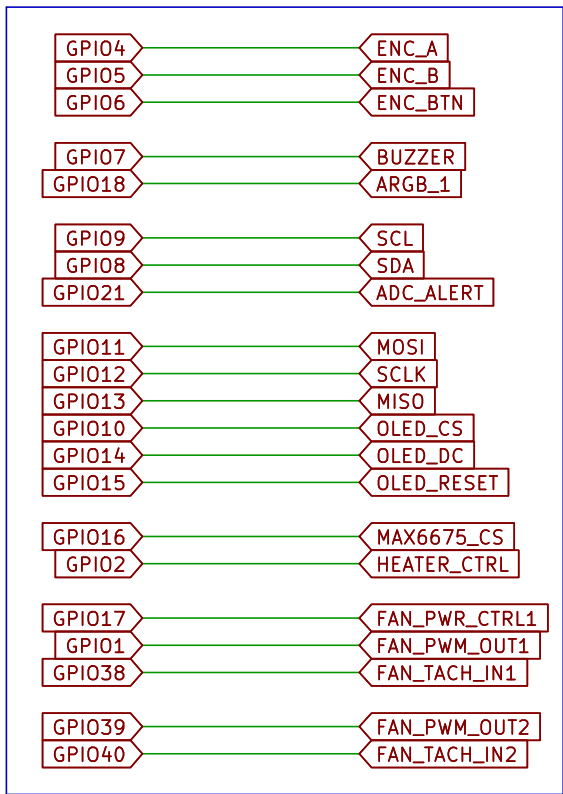
Title: ReflowDesk AT-MK1 (Power Supply)

Size: A3	Date: 2026-04-30	Rev: v1.0
KiCad E.D.A. 10.0.1		Id: 3/6

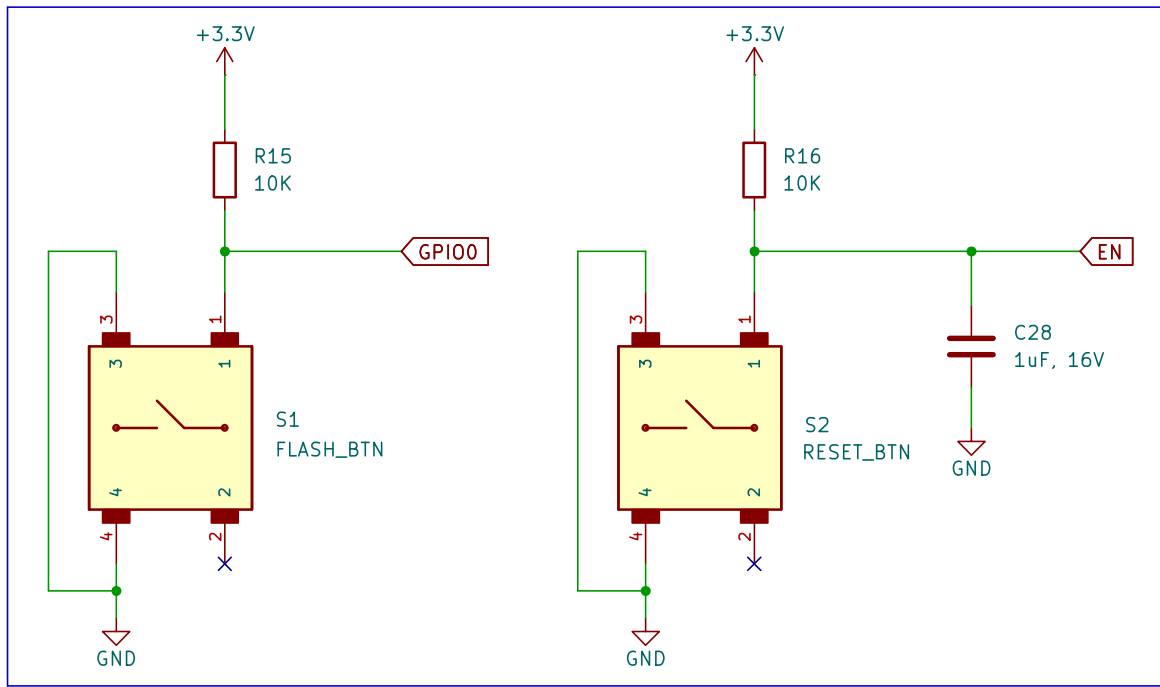
32-bit MCU (Flash: 8MB, PSRAM: 8MB)



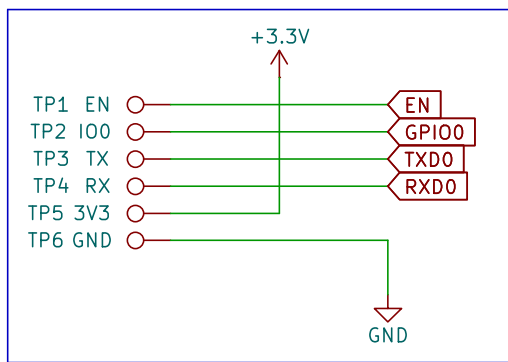
MCU GPIO Pins Allocation/Configuration



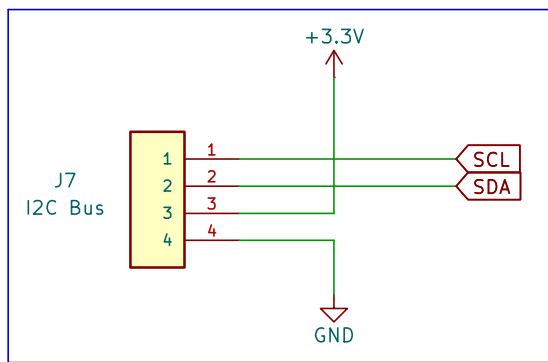
MCU Flash and Reset Buttons



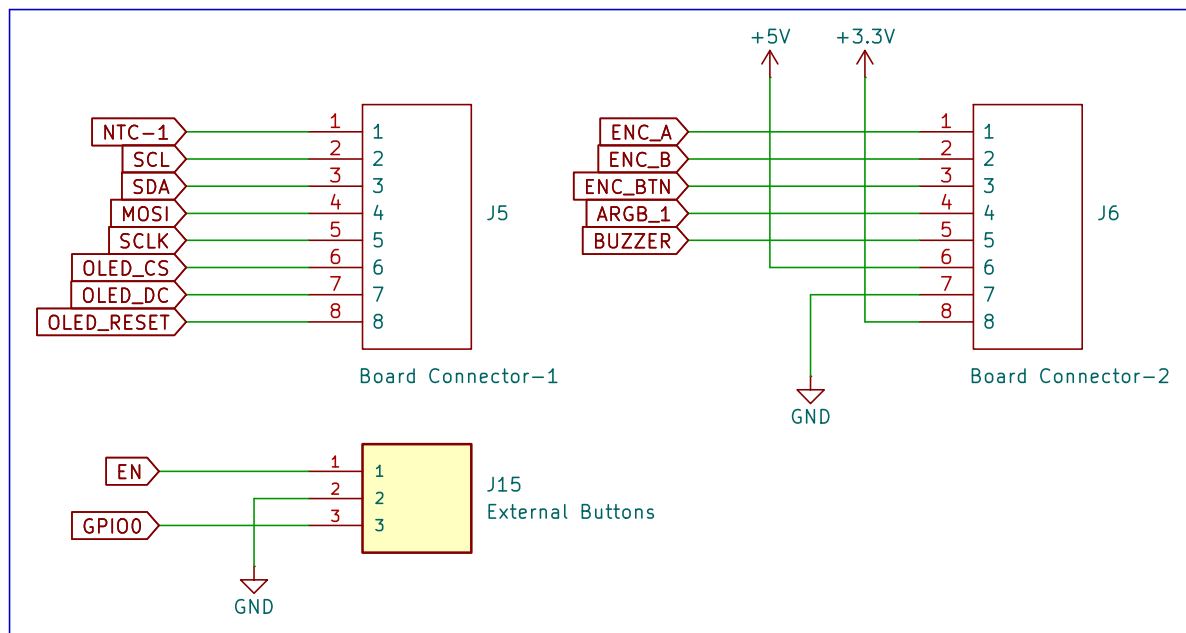
UART Test Pads



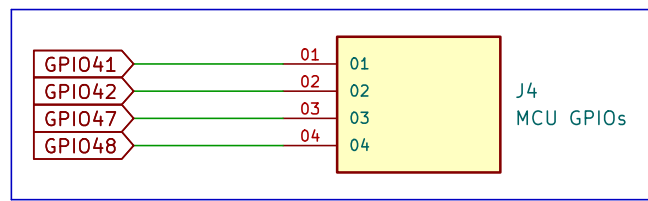
I2C Bus Connector



DaughterBoard (User Interface Board) Connectors



MCU GPIO Breakout (Future Usage)



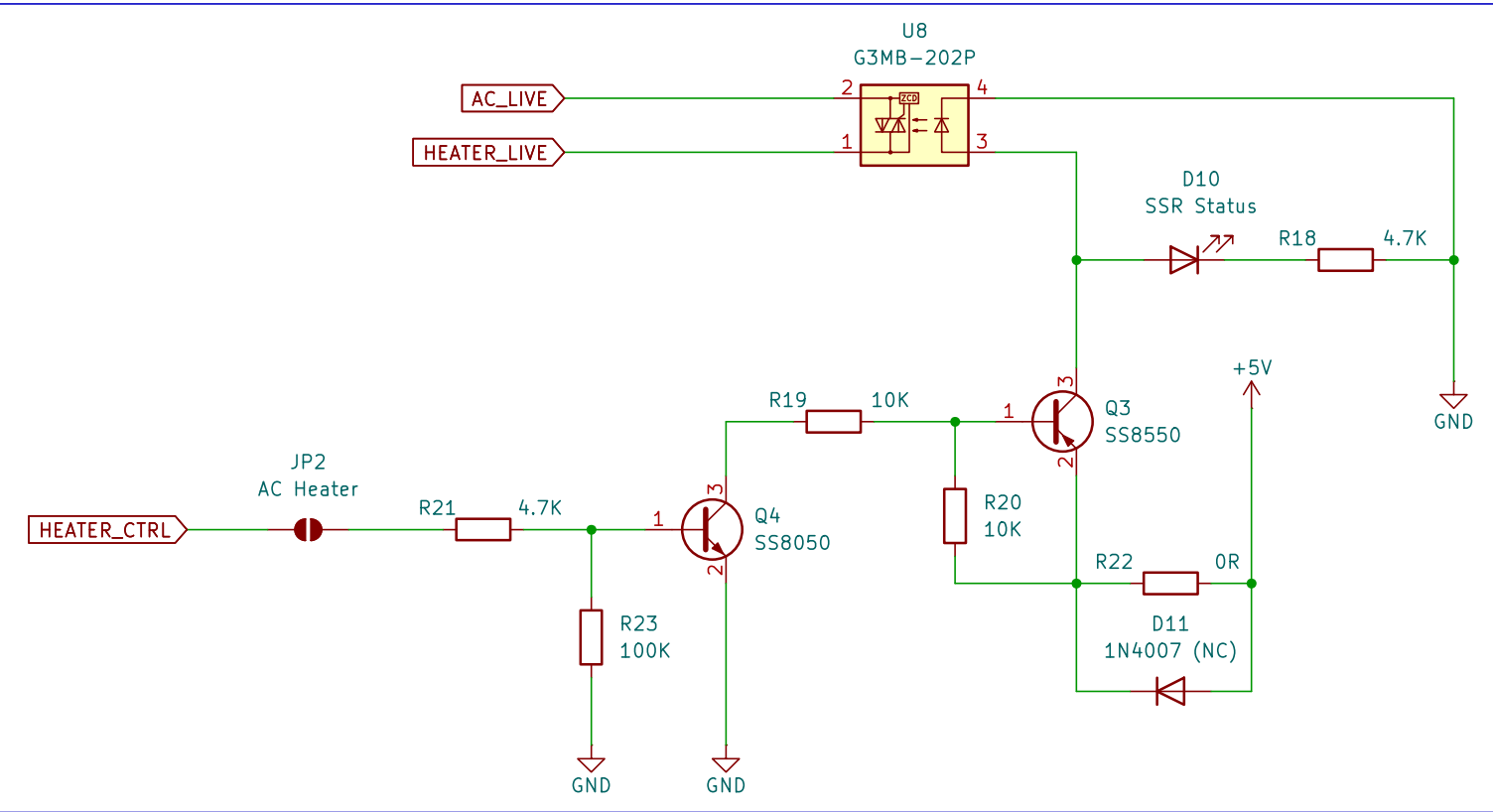
SMD Reflow Soldering Hot Plate
A-Tech Officials
Sheet: /MCU/
File: mcu.kicad_sch

Title: ReflowDesk AT-MK1 (MCU)

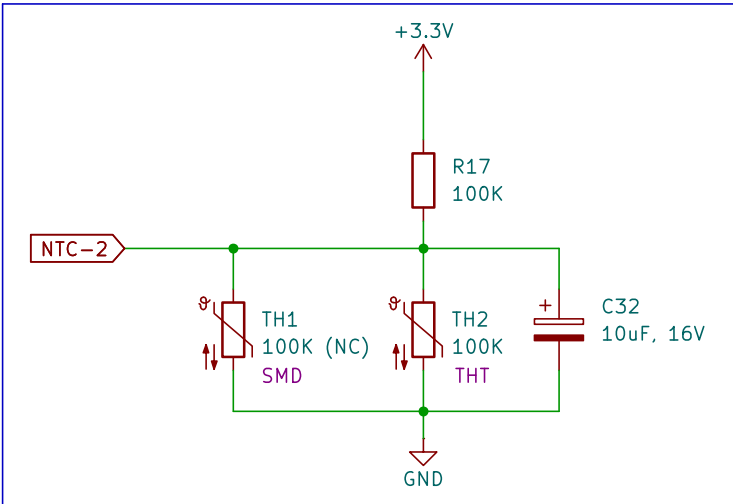
Size: A3
Date: 2026-04-30
KiCad E.D.A. 10.0.1

Rev: v1.0
Id: 4/6

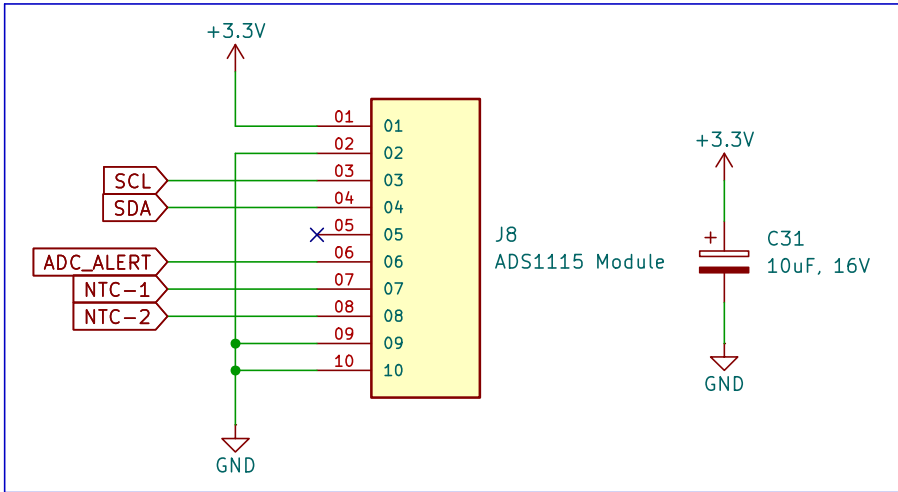
220VAC PTC Heating Element Control (Active-Low SSR, Active-High Signal)



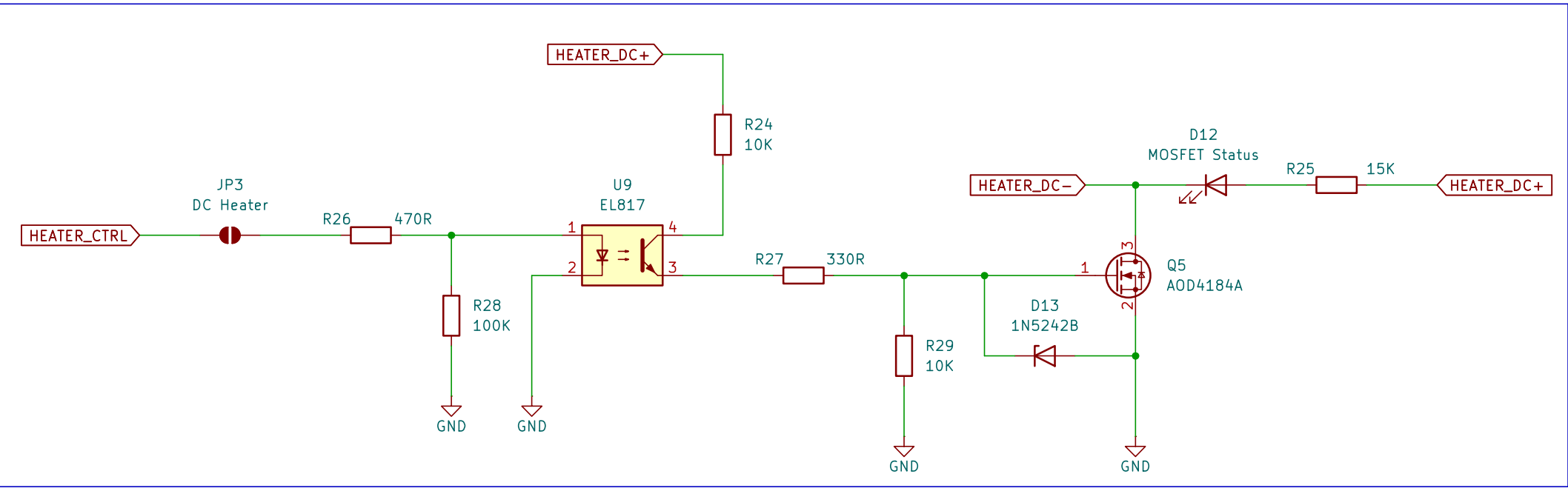
NTC Ambient Temp Sensor (Controller Board)



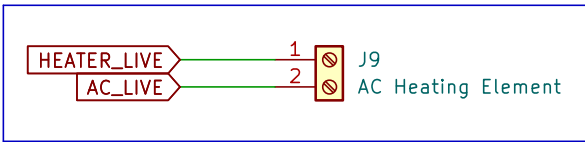
ADS1115 16-bit ADC Module Connector



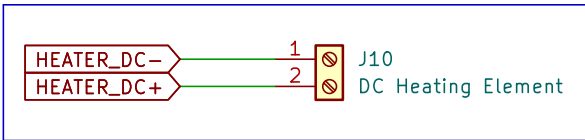
DC PTC Heating Element Control (Low-Side MOSFET Switch)



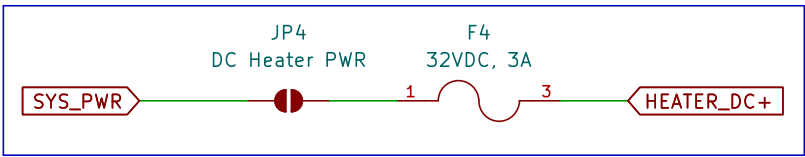
220VAC PTC Heating Element Connector



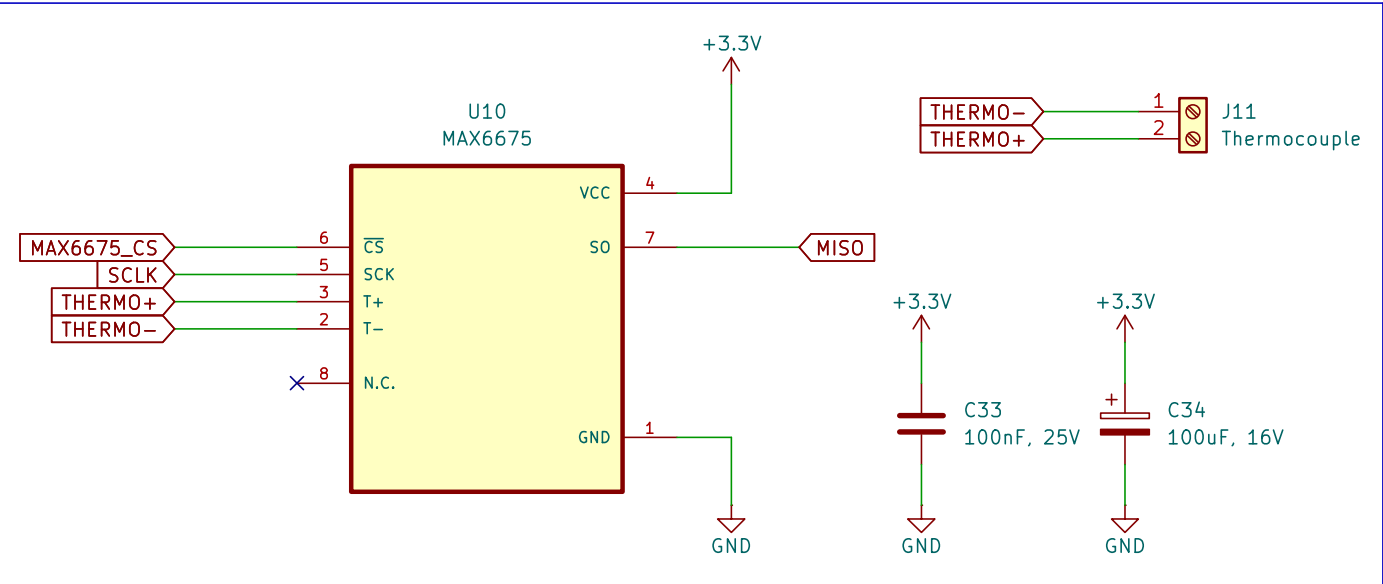
DC PTC Heating Element Connector



DC PTC Heating Element Power Enable Jumper



Hot Plate Temperature Sensor (K-Type Thermocouple)



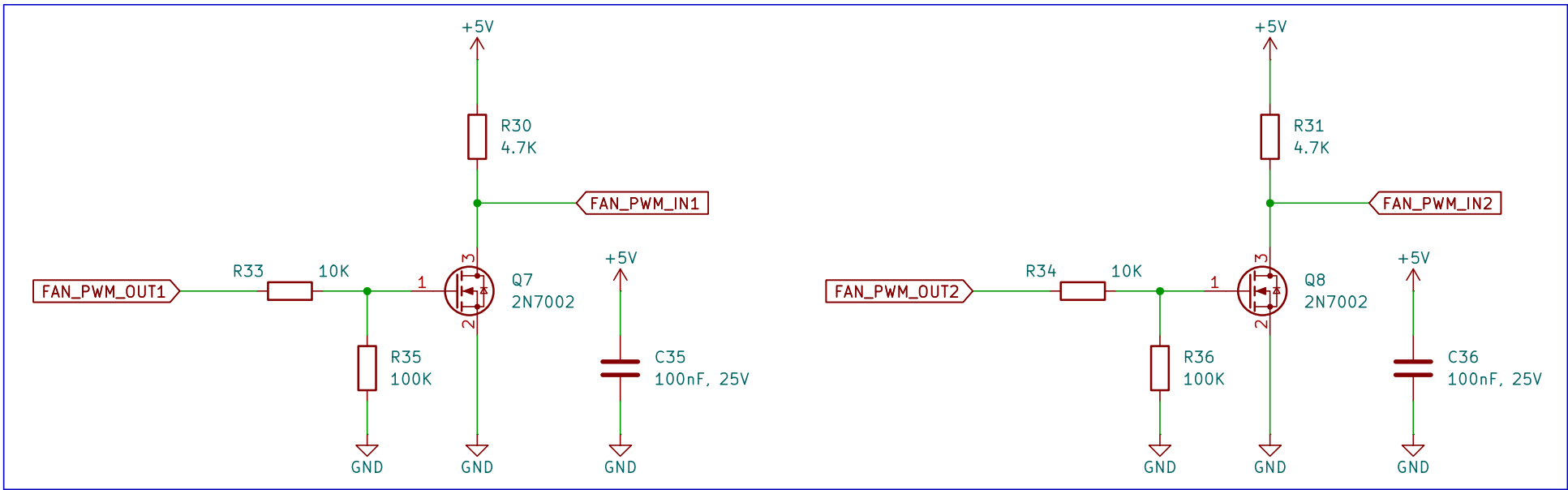
SMD Reflow Soldering Hot Plate
A-Tech Officials
Sheet: /Heating Element/
File: heating_element.kicad_sch

Title: ReflowDesk AT-MK1 (Heating Element)

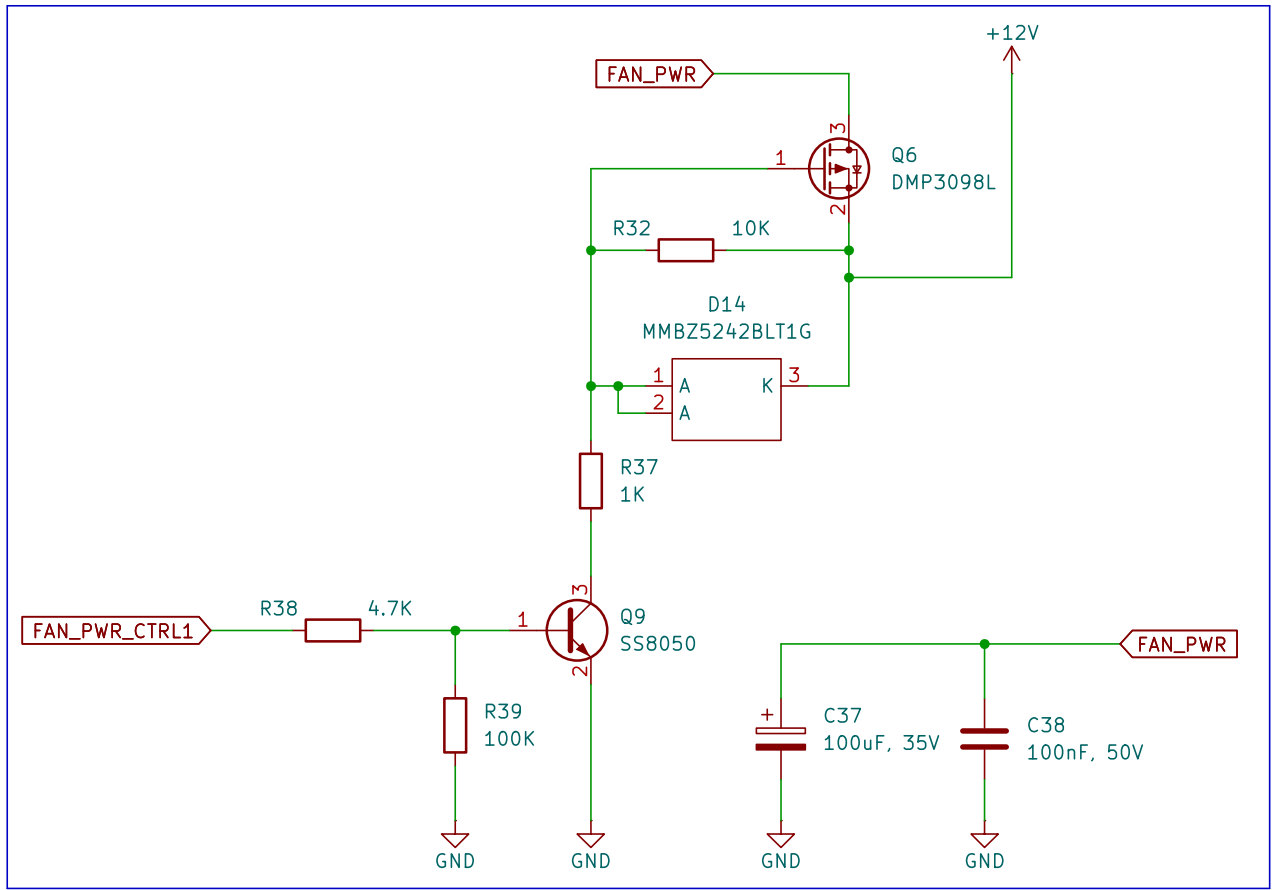
Size: A3 Date: 2026-04-30
KiCad E.D.A. 10.0.1

Rev: v1.0
Id: 6/6

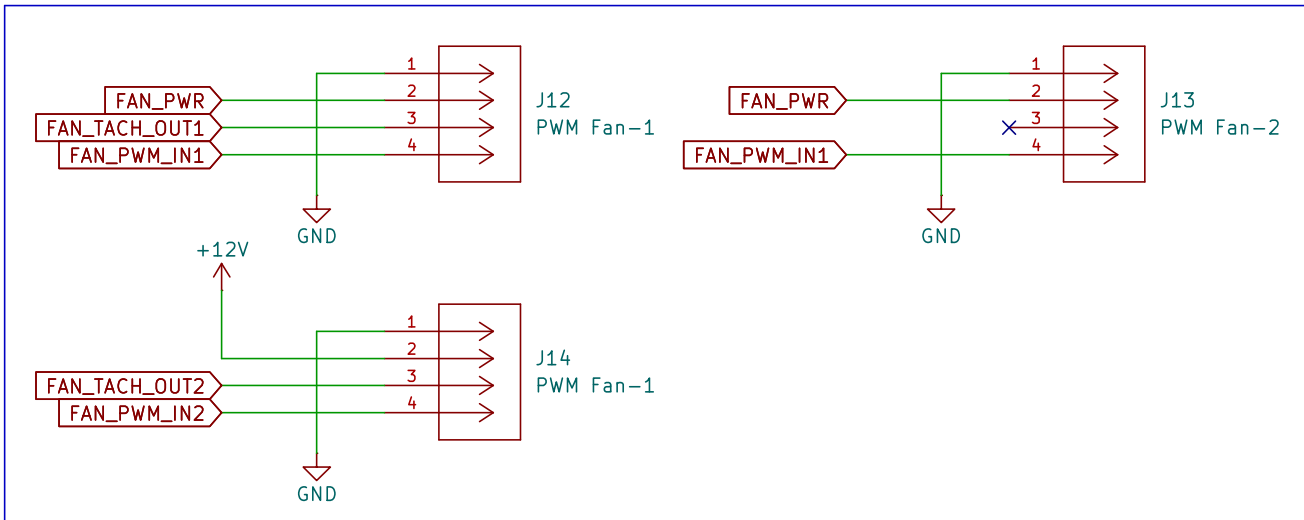
N-MOSFET Open-Drain Level Shifter / PWM Distribution Buffer



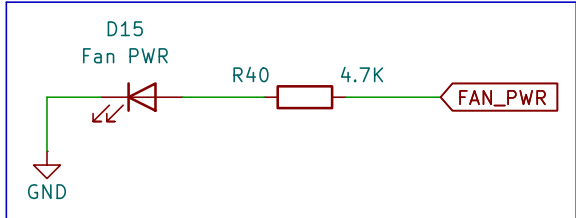
12VDC PWM Fan Power Control



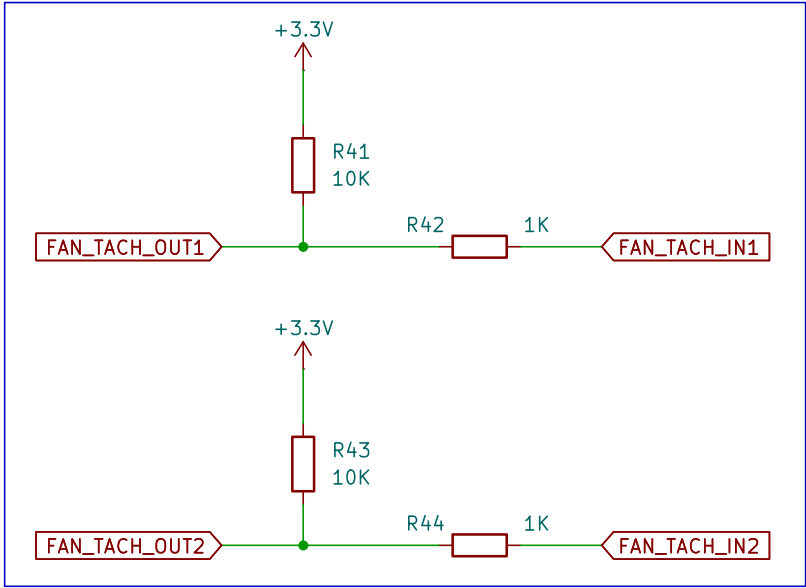
12VDC PWM Fan Connectors



Fan Power Status LED



Fan Tachometer Read Circuit



SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /Cooling System/

File: cooling_system.kicad_sch

Title: ReflowDesk AT-MK1 (Cooling System)

Size: A3

Date: 2026-04-30

Rev: v1.0

KiCad E.D.A. 10.0.1

Id: 7/6